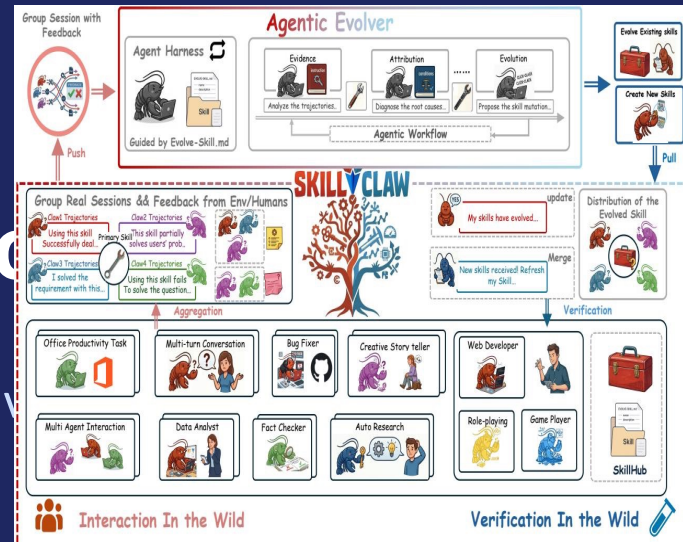


SkillClaw: Collective Skill Multi-User LLM Agent Eco

Let Skills Evolve Collectively with an Agentic Evolution



arXiv:2604.08377

Agent Skills Remain Static After Deployment — Users Rediscover Solutions Independently

Static Skill Paradigm

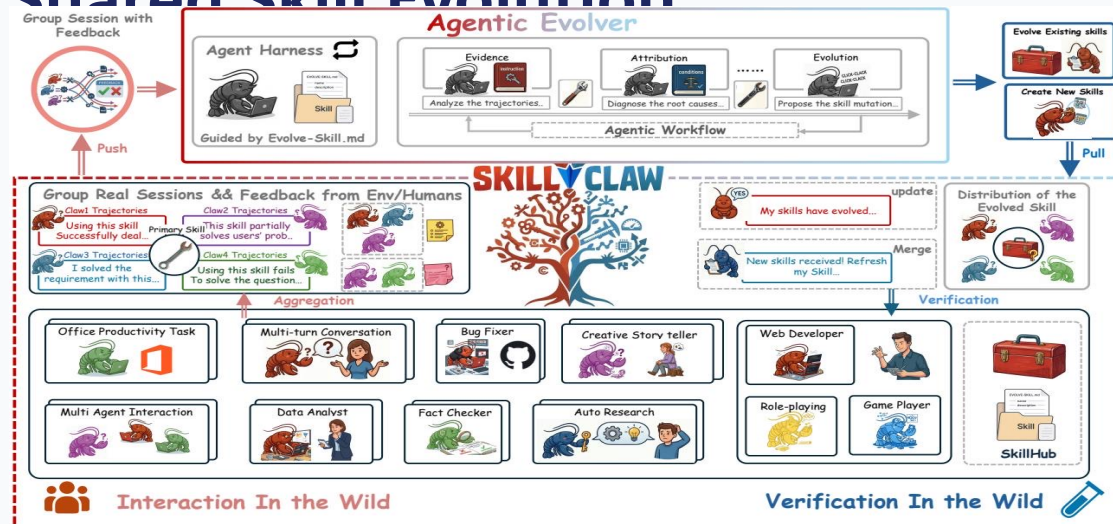
- Skills manually installed from a centralized hub
- Solutions discovered during interaction rarely persist
- Same failure/recovery patterns repeated across users
- Memory-based methods tied to specific instances
- Skill libraries treated as static resources

Evolving Skill Paradigm

- Skills improve continuously from runtime experience
- Cross-user trajectories aggregated into shared evidence
- System-level knowledge accumulation over time
- Agentic reasoning drives open-ended skill mutations
- Verified updates synchronized to all agents

Key Insight: Each user is forced to rediscover solutions independently, preventing knowledge from accumulating at the system level.

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution



Evidence

Analyze trajectories to identify behavioral patterns

Attribution

Diagnose root causes of failures and successes

Evolution

Propose skill mutations — refine or create new skills

Distribution

Verify, merge into shared repo, and sync to

Multi-user Interaction → Session Collection → Skill Evolution → Skill Synchronization

Independent agents produce structured session trajectories preserving full action–feedback causal chains. Trajectories are aggregated across users, grouped by referenced skills, and fed into an autonomous agentic evolver performing open-ended reasoning over interaction evidence.

Agentic Evolver: Evidence → Attribution → Evolution in Three Stages

01 Evidence

Analyze the aggregated trajectories to identify behavioral patterns. The evolver examines action–feedback causal chains across diverse user contexts to surface recurring success signals and failure modes.

02 Attribution

Diagnose root causes of failures and successes. By cross-referencing complementary views from different users, the evolver separates generalizable conditions from idiosyncratic fixes.

03 Evolution

Propose skill mutations — refining existing skills or creating new ones. Updates are produced through open-ended agent reasoning, not predefined update rules, enabling context-aware adaptation.

The evolution process is driven by an autonomous agent performing open-ended reasoning over interaction evidence, not predefined update rules.

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Why Cross-User Aggregation Matters

Different users exercising the same skill under diverse contexts produce complementary views of that skill's behavioral boundary.

Reveals both conditions under which the skill works and those under which it breaks.

A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix.

Compatible Claw-Style Systems

● OpenClaw

● CoPaw

● IronClaw

● PicoClaw

● ZeroClaw

● NanoClaw

● NemoClaw

Evaluation on WildClawBench (Qwen3-Max backbone) demonstrates substantial improvements across tasks.

Case Study: Multimodal Creative Pipeline Skill — 6 Days of Evolution Attempts

Day	Candidate Skill	Validator	Next-Day Action
1	validate-tmp-workspace-inputs	Accept	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Reject	Keep current best pool
3	multimodal-creative-task-pipeline	Reject	Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	Reject	Keep current best pool
5	creative pipeline + per-file fail-fast validation	Reject	Keep current best pool
6	extended PDF-to-poster / document-to-visual paths	Reject	Not admitted to next cycle

Only 1 of 6 candidates was accepted, illustrating the conservative verification process that prevents regressions.

Case Study: Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

Day	Candidate Skill	Validator	Next-Day Action
1	git-push-with-auth-fallback	Accept	Add to Safety best pool
2	git-push-with-auth-fallback	Accept	Keep updated best pool
3	git-push-with-auth-fallback + git-clone-to-directory	Accept	Keep current best pool
4	(none) — same-pool retest	Accept	Continue same best pool
5	git-push-with-auth-fallback	Reject	Keep current best pool
6	git-push-with-auth-fallback	Reject	Not admitted to next cycle

3 of 6 days yielded accepted updates, showing successful iterative refinement before reaching a plateau.

Key Takeaways: Three Properties That Distinguish SkillClaw from Existing Systems

Collective Evolution

1

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem. Cross-user trajectory aggregation provides complementary behavioral signals that a single user cannot generate alone.

Fully Automatic

2

Skill evolution is driven by runtime interaction without manual curation or explicit user intervention. The system improves as a side effect of normal usage.

Agentic Evolution Paradigm

3

Skill updates are produced through open-ended reasoning over interaction evidence rather than predefined update rules, enabling context-aware mutations that rigid heuristics cannot match.

Implication: Continuously improving agent systems are feasible when skills evolve collectively, automatically, and agentially.